

STUDY GUIDE

Questions 1-65 - Typed from the supplied study pages

Answers are bold and highlighted. Items 13, 14, and 60 refer to figures that were not visible in the photos.

Motor Control & Networks

1. In a stepper motor, what is the maximum torque the motor can withstand without moving from its position with the stator energized?
Answer: **HOLDING torque** [MC3L7]
2. In a stepper motor, what is the maximum torque the motor can withstand without moving from its position with the stator de-energized?
Answer: **DETENT torque** [MC3L7]
3. When the system is more capacitive, how is the phase of the current related to the voltage, and what power factor does the system have?
Answer: **The current is OUT OF PHASE WITH the voltage, and the system has a LAGGING power factor.** [MC3L7]
4. Which type of motor must have a position feedback sensor?
Answer: **SERVO MOTORS** [MC3L8]
5. In what network topology are all machines connected to a central network media (cable) called the backbone?
Answer: **BUS topology** [MC3L8]
6. In what network topology are machines interconnected in the shape of a closed loop?
Answer: **RING topology** [MC3L8]
7. Which type of control network permits all nodes on the network to simultaneously access the same data from a single source?
Answer: **CONTROL-NET** [MC3L8]
8. Which control network topology has all network devices connected to a central hub or switch?
Answer: **STAR TOPOLOGY** [MC3L8]
9. In the control system block diagram, which lines represent a divergent control path?
Answer: **5, 6, and 7** [MC3L9]
10. In the control system block diagram, which line represents a feedback control path?
Answer: **3** [MC3L9]
11. In the motor control troubleshooting steps, where does the missing "Correct the problem" step go?
Answer: **Between DECIDE WHAT TYPE OF TEST TO PERFORM and VERIFY CORRECT OPERATION.** [MC3L9]
12. When troubleshooting a motor control system, information funneling involves how many levels of testing?
Answer: **4 LEVELS OF TESTING** [MC3L9]

Hazardous Locations

13. Hazardous Location 9 - Question/Picture 1
Answer: **Reference only - the actual picture/question is not visible in the supplied photo.** [Haz Location 9]
14. Hazardous Location 9 - Question/Picture 2
Answer: **Reference only - the actual picture/question is not visible in the supplied photo.** [Haz Location 9]
15. To minimize the effects of "pressure piling," how far apart should seals generally be inserted in long conduit runs?
Answer: **Not more than 50-100 ft apart, depending on conduit size.** [HzL9]
16. Where ventilation is not provided in a minor repair garage, what floor area is considered Division 2, Zone 2?
Answer: **The area up to 18 in. above the floor, EXTENDING 20 ft HORIZONTALLY in all directions.** [NEC 514.3]

17. What is vapor density?

Answer: **The weight of a volume of pure vapor or gas compared with the weight of an equal volume of dry air at the same temperature and pressure.** [HzL9]

Security & Access Control

18. What is a partition in a security system?

Answer: **A PARTITION is a portion of the security system that operates under a different program than the rest of the system.** [BA2L4]

19. What type of security system circuit uses devices connected in series?

Answer: **OPEN LOOP (as highlighted in the source; a handwritten note beside it questions "closed loop?")** [BA2L4]

20. An open-loop security system circuit uses devices that are normally what when there is no abnormal condition present?

Answer: **OPEN** [BA2L4]

21. What type of system is typically used in high-security buildings and continuously occupied facilities such as hospitals, universities, airports, and industrial plants?

Answer: **A SUPERVISORY SYSTEM** [BA2L4]

22. What type of protection monitors one particular object?

Answer: **SPOT PROTECTION** [BA2L4]

23. What type of protection monitors only one defined location?

Answer: **SPECIFIC AREA PROTECTION** [BA2L4]

24. What type of security protection monitors specific points that might allow entry to or exit from a secure area?

Answer: **POINT OF ENTRY** [BA2L4]

25. What credential technology requires a combination of two or more credentials in a single unit before allowing access?

Answer: **MIXED CREDENTIAL technology** [BA2L5]

26. What may an access control system use to store an access-control database and confirm a match with stored information?

Answer: **A COMPUTER** [BA2C9 packet]

27. What process involves collecting information from the user and verifying credentials?

Answer: **AUTHENTICATION** [BA2L5]

Communications, CCTV & Paging

28. What device modulates an analog carrier signal into digital information for transmission over a medium?

Answer: **A MODEM** [BA2L6]

29. What signal can be used to remotely control building systems over dial-up lines?

Answer: **A DTMF signal** [BA2L6]

30. What device can record up to 16 cameras simultaneously?

Answer: **A MULTIPLEX DVR** [BA2L6]

31. What is the maximum number of pixels that can be displayed on a screen called?

Answer: **DISPLAY RESOLUTION** [BA2L6]

32. What is the main purpose of installing telephone line-seizure devices upstream of other telephone devices?

Answer: **So that, if necessary, an emergency control panel can take control of the telephone line.** [BA2L6]

33. What type of camera can be remotely controlled to change its field of view?

Answer: **PAN-TILT-ZOOM (PTZ) camera** [BA2L6]

34. What is a common type of paging system used in commercial settings?

Answer: **CENTRAL AMPLIFIED SYSTEM** [BA2L6]

35. Which item is NOT a type of digital monitor used in a CCTV system?

Answer: **RESISTIVE TOUCH-SCREEN CRT** [BA2L6]

Elevators

36. What type of elevator uses a worm-gear reduction unit that allows precise control of elevator position?
Answer: **A GEARED TRACTION elevator** [BA2L7]
37. What safety mechanism activates the emergency brakes on an elevator?
Answer: **The GOVERNOR** [BA2L7]
38. What elevator mode allows emergency personnel to manually operate an elevator that is in Phase One mode?
Answer: **PHASE TWO MODE** [BA2L7]
39. Which elevator mode releases elevator control from the central controller?
Answer: **INDEPENDENT SERVICE MODE** [BA2L7]
40. What are reasons to put an elevator into a special operating mode?
Answer: **CONVENIENCE, EFFICIENCY, SAFETY, and SERVICE** [BA2L7]
41. What elevator safety feature shuts the elevator down when water from a sprinkler system could make safety features, such as braking, useless?
Answer: **ELEVATOR SHUT DOWN (SHUNT TRIP)** [BA2L7]

PLC Troubleshooting & Maintenance

42. What are the two categories of power-supply problems for PLCs?
Answer: **Loss of power and LOW VOLTAGE** [IPLC-11]
43. When a control transformer is fully loaded, what condition indicates that it is overloaded?
Answer: **A VOLTAGE DROP OF MORE THAN 5%** [IPLC-11]
44. When troubleshooting PLC hardware, what is the first step in testing a circuit?
Answer: **TURN POWER OFF, LOTO** [IPLC-11]
45. Forcing a PLC during what process verifies that the wiring to input devices and the action are correct?
Answer: **TROUBLESHOOTING** [IPLC-12]
46. What is the first step in troubleshooting a PLC application?
Answer: **VIEW THE PROGRAM** [IPLC-12]
47. When an input terminal of a PLC is forced, how does the LED display rectangle for that input terminal appear?
Answer: **It APPEARS EMPTY.** [IPLC-12]
48. When using PLC programming software, which key generates a help window for the selected dialogue box, instruction, view, or window?
Answer: **F1** [IPLC-12]
49. When using the TND instruction for PLCs, where does the debugging process typically start?
Answer: **AT THE BEGINNING OF A PROGRAM** [IPLC-12]
50. What are the small windows in PLC software that allow users to make choices and enter information called?
Answer: **DIALOGUE BOXES** [IPLC-12]
51. What PLC state of operation has the operating cycle stopped while the CPU remains ready to accept information?
Answer: **PROGRAM MODE** [IPLC-13]
52. Why may PLC hardware maintenance require the assistance of a second technician or operator?
Answer: **Because SOME PLC APPLICATIONS MUST BE STOPPED IN A SPECIFIC SEQUENCE.** [IPLC-13]
53. During energized PLC maintenance, which lights must be lit?
Answer: **The LED POWER and RUN lights** [IPLC-13]
54. How often is PLC system maintenance typically performed?
Answer: **ANNUALLY** [IPLC-13]
55. What does visual PLC maintenance cover?
Answer: **The PLC enclosure, PLC-related components, PLC INPUT DEVICES, and OUTPUT COMPONENTS.** [IPLC-13]

56. What is RUN MODE?

Answer: **A PLC state of operation where the operating cycle of the PLC is running continuously to control a machine or process.** [IPLC-13]

57. What is software and program verification?

Answer: **A process that ensures a PLC can be accessed by a PC and that the PC has a current copy of the program in the PLC.** [IPLC-13]

58. What do instruction comments provide?

Answer: **Information about a specific instruction associated with a specific address.** [IPLC-13]

NEC Selective Coordination & GFPE

59. According to the NEC, what is selective coordination?

Answer: **The localization of an overcurrent condition to restrict outages to the circuit or equipment affected, accomplished by the selection and installation of OCPDs and their ratings or settings.** [CSP6L7]

60. Using the figure, determine which OCPDs OCPD #2 must be selectively coordinated with.

Answer: **The figure/answer is not visible in the supplied photo. Reference shown: NEC 700.32.** [CSP6L7 / NEC 700.32]

61. Why does the NEC require selective coordination?

Answer: **FOR SYSTEMS OR CIRCUITS IN WHICH THE LOADS ARE VITAL FOR LIFE SAFETY AND PUBLIC SAFETY.** [CSP6L7]

62. According to NEC 230.95, on what solidly grounded wye electric services is ground-fault protection of equipment (GFPE) required?

Answer: **More than 150 V to ground, but not exceeding 1,000 V phase-to-phase. Handwritten example: 480/277 V wye system.** [CSP6L8 / NEC 230.95]

63. Where is ground-fault protection of equipment not permitted or not required under the noted exception?

Answer: **It is not permitted for FIRE PUMP CIRCUITS and is not required for a continuous industrial process where a non-orderly shutdown would introduce additional or increased hazards.** [CSP6L8 / NEC 230.95 Exception]

64. When do ground-fault protection requirements NOT apply to a service disconnect?

Answer: **When it is used for a continuous industrial process where a non-orderly shutdown would introduce additional or increased hazards.** [CSP6L8]

65. What are the two common methods used for ground-fault protection of equipment?

Answer: **ZERO-SEQUENCE METHOD and RESIDUAL METHOD** [CSP6L8]